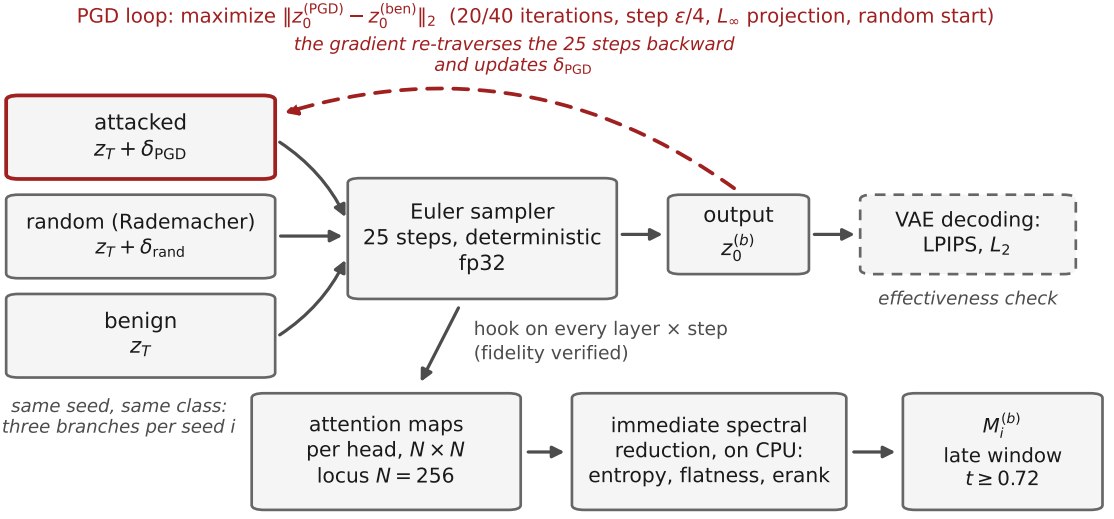
PGD loop: maximize $\|z_0^{(\mathrm{PGD})} - z_0^{(\mathrm{ben})}\|_2$ (20/40 iterations, step $\varepsilon/4$, L_∞ projection, random start)
the gradient re-traverses the 25 steps backward
and updates δ_{PGD}
attacked
$z_T + \delta_{\mathrm{PGD}}$
random (Rademacher)
$z_T + \delta_{\mathrm{rand}}$
benign
z_T
Euler sampler
25 steps, deterministic
fp32
output
$z_0^{(b)}$
VAE decoding:
LPIPS, L_2
effectiveness check
hook on every layer × step
(fidelity verified)
same seed, same class:
three branches per seed i
attention maps
per head, $N \times N$
locus $N = 256$
immediate spectral
reduction, on CPU:
entropy, flatness, erank
$M_i^{(b)}$
late window
$t \geq 0.72$